

SIMATS ENGINEERING
Slot: C, Course Code: ITA0501, Course Name: Computer Vision

Duration: 60 mins

Marks: 25

Assessment Tool 2: Industry Problem Solving Task (Medium)

Assessment Tool Weightage: 20%

COURSE OUTCOME COVERED

CO4: Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)

Q. No.	Questions
1	Smart Surveillance System Selection A city surveillance system must operate under the following conditions: If latency > 100 ms → system fails • If accuracy < 85% → unacceptable • Hardware supports only moderate computation Models available: Viola-Jones: 78%, low latency, YOLO: 92%, moderate latency, Deep Learning: 95%, high latency Compare the models and evaluate which satisfies all constraints. Justify your decision using logical reasoning based on given conditions.
2	Autonomous Vehicle Detection Logic A vehicle detection system must: Prioritize safety (accuracy > 90%). Maintain response time < 50 ms Options: Method A: 88% accuracy, 30 ms. Method B: 93% accuracy, 70 ms. Method C: 91% accuracy, 45 ms Compare the methods and evaluate which should be selected. Justify your choice using constraint-based reasoning.
3	Motion Analysis Decision Problem A traffic system requires: Accurate motion direction detection, Robustness in dense traffic Results: Optical Flow: accurate but noisy in dense scenes, Motion Estimation: stable but less detailed Compare both approaches and evaluate which is suitable under high-density conditions. Justify using scenario-based logic.
4	Background Modeling Evaluation A GMM-based system produces: Static scenes: 90% accuracy, Dynamic scenes: 65% accuracy Requirement: System must maintain ≥80% accuracy across all scenarios Compare performance and evaluate whether GMM alone meets requirements. Justify logically and suggest a decision.
5	Depth Estimation Strategy Constraints: Only one camera available. Moderate depth accuracy acceptable Options: Stereo Vision: high accuracy, needs two cameras, Structure from Motion: moderate accuracy, single camera Compare both approaches and evaluate which is feasible. Justify using constraint-based reasoning.

Code of Conduct:

I SOMISETTY VENKATA SAI NAVEEN KUMAR (192425176) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student
SOMISETTY VENKATA SAI NAVEEN KUMAR
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RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	MARKS
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding, some requirements unclear	Poor understanding, misinterprets problem context	
Application of Engineering Concept	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts	
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution	
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools	
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis	
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation	

Signature of the Course Faculty
Dr P Lourdu Mahimai Doss

Answers – Industry Problem Solving Task

1. Smart Surveillance System Selection

Given constraints:

- Latency > 100 ms $\rightarrow$ system fails.
- Accuracy $< 85\%$ $\rightarrow$ unacceptable.
- Hardware supports only moderate computation.

Model	Accuracy	Latency/Computation	Decision
Viola-Jones	78%	Low latency	Accuracy unacceptable
YOLO	92%	Moderate latency	Suitable
Deep Learning	95%	High latency	Too computationally expensive

Answer: YOLO should be selected.

Viola-Jones has low latency, but its accuracy of 78% is below the required 85%, so it cannot be used. Deep Learning provides the highest accuracy of 95%, but its high latency and computational requirements conflict with the hardware limitation. YOLO provides 92% accuracy with moderate computational requirements and satisfies the accuracy and latency constraints.

Conclusion: YOLO provides the best balance between **accuracy, latency, and computational cost**, making it the most suitable model for the surveillance system.

2. Autonomous Vehicle Detection Logic

Requirements:

- Accuracy must be $> 90\%$.
- Response time must be < 50 ms.
- Safety is the priority.

Method	Accuracy	Response Time	Meets Requirements?
Method A	88%	30 ms	Accuracy fails
Method B	93%	70 ms	Response time fails
Method C	91%	45 ms	Both satisfied

Answer: Method C should be selected.

Method A has a fast response time of 30 ms, but its accuracy is only 88%, which is below the required 90%. Method B provides 93% accuracy but has a response time of 70 ms, exceeding the 50 ms limit. Method C achieves 91% accuracy and responds within 45 ms.

Therefore, **Method C is the only method satisfying both constraints.**

Conclusion: Method C is the safest and most appropriate choice because it meets both the accuracy and response-time requirements.

3. Motion Analysis Decision Problem

Requirements:

- Accurate motion-direction detection.
- Robustness in dense traffic.

Optical Flow:

- Provides accurate motion information.
- Can capture detailed motion and direction.
- Becomes noisy when many objects are moving simultaneously in dense traffic.

Motion Estimation:

- Produces more stable motion information.
- Less affected by crowded scenes.
- Provides less detailed motion information.

Answer: Motion Estimation should be selected for high-density traffic conditions.

In dense traffic, many vehicles and objects move simultaneously. Optical Flow may generate noisy vectors because of overlapping movements and complex motion patterns. Motion Estimation provides more stable results and is therefore more robust in a crowded environment.

However, if the application mainly requires detection of **small or subtle movements**, Optical Flow would be advantageous because it provides more detailed motion information.

Conclusion: For the given scenario, where **robustness in dense traffic** is important, **Motion Estimation is the better choice**. A hybrid system could also use Motion Estimation for general traffic and Optical Flow for detailed motion analysis when required.

4. Background Modeling Evaluation

Given:

- GMM accuracy in static scenes = **90%**
- GMM accuracy in dynamic scenes = **65%**
- Required accuracy = **more than 80% across all scenarios**

The GMM performs well in static environments because the background remains relatively stable. Its accuracy is 90%, which satisfies the requirement.

However, in dynamic scenes, the accuracy falls to 65%.

Therefore, GMM **does not satisfy the requirement across all scenarios**.

Suggested design

A better system should use **adaptive background modeling**. The model should continuously update its background representation according to environmental changes.

Possible improvements include:

- Adaptive learning rate.
- Shadow detection and removal.
- Morphological noise filtering.
- Temporal consistency checking.

- Combining background subtraction with an object detector such as YOLO.

Conclusion: GMM alone is **not sufficient** because its dynamic-scene accuracy is only 65%. An adaptive or hybrid background-modeling system should be used to improve performance in dynamic environments and achieve the required **>80% accuracy**.

5. Depth Estimation Strategy

Constraints:

- Only **one camera** is available.
- Moderate depth accuracy is acceptable.

Approach	Accuracy	Camera Requirement	Suitable?
Stereo Vision	High	Two cameras	Not feasible
Structure from Motion	Moderate	Single camera	Feasible

Answer: Structure from Motion (SfM) should be selected.

Stereo Vision generally provides accurate depth estimation by calculating disparity between images captured from two different viewpoints. However, it requires two cameras, which violates the given constraint.

Structure from Motion can estimate depth using a **single camera** by analyzing changes in the scene across multiple frames as the camera moves. Although its depth accuracy may be lower than stereo vision, moderate accuracy is acceptable for this application.